GRADE 10 · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 02

Third Term Test (Full Syllabus)

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Time: 3 hours **Total marks:** 100

Name: ..... Class: ..... Date: .....

Instructions:

- This question paper consists of two parts, **A** and **B**. Answer **all** the questions.
- Answer the 20 questions in **Part A** in the space provided on this paper itself. (2 marks for each question)
- Answer the 6 questions in **Part B** on the separate sheets of paper provided. (10 marks for each question)
- Show the steps of your calculations clearly. Draw graphs on the graph paper provided.
- Where necessary, take the value of π as $\frac{22}{7}$. The use of calculators is not allowed.

Part	A	B1	B2	B3	B4	B5	B6	Total
Marks	/40	/10	/10	/10	/10	/10	/10	/100

Part A

Answer all the questions in the space provided on this paper. (2 × 20 = 40 marks)

1. Find the square root of 5.29.

Answer: .....

2. Find the value of: $(\frac{3}{5} \times 2\frac{1}{2}) - (\frac{1}{4} \div \frac{1}{2})$

Answer: .....

3. Expand: $(x + 4)(x - 7)$

Answer: .....

4. Factorise: $9a^2 - 16b^2$

Answer: .....

5. Factorise completely: $2x^2 - 8x - 42$

Answer: .....

6. Simplify: $\frac{x^2 - 9}{x^2 + 5x + 6}$

Answer: .....

7. Find the area of a sector of a circle of radius 14 cm and centre angle 90°.

Answer: .....

8. An exterior angle of a triangle is 125°. If one of the two interior opposite angles is 48°, find the other angle.

Answer: .....

9. Find the value of: $\lg 50 + \lg 4 - \lg 2$

Answer: .....

10. If lg 2 = 0.3010, write the value of lg 0.8 in the form used in logarithm tables (with the decimal part positive).

Answer: .....

11. Which term of the arithmetic progression 3, 7, 11, ... is 87?

Answer: .....

12. If ξ = {integers from 1 to 10}, A = {even numbers} and B = {multiples of 3}, write the set $A' \cap B$ by listing its elements.

Answer: .....

13. When a sum of Rs. 40 000 is deposited for 2 years at an annual compound interest rate of 10%, find the total amount received at the end.

Answer: .....

14. A can complete a piece of work in 6 days and B can complete it in 12 days. Find the number of days taken to complete the work if both work together.

Answer: .....

15. Find the average speed of a vehicle that travels a distance of 180 km in 2 hours 15 minutes.

Answer: .....

16. Make l the subject of the formula $S = \frac{n}{2}(a + l)$.

Answer: .....

17. When two fair coins are tossed together, find the probability of getting at least one head.

Answer: .....

18. In the parallelogram ABCD, AB = (2x + 3) cm and CD = (15 − x) cm. Find the length of AB.

Answer: .....

19. The area of a parallelogram standing on a given base and between a pair of parallel lines is 48 cm^2. Find the area of a triangle standing on the same base and between the same pair of parallel lines.

Answer: .....

20. On a map drawn to the scale 1 : 50 000, find, in kilometres, the actual distance represented by 3.5 cm.

Answer: .....

Part B

Answer all the questions on separate sheets of paper. (10 × 6 = 60 marks)

1. (a) Simplify: $\frac{2}{x+3} + \frac{3}{x-3} - \frac{18}{x^2-9}$ (4 marks)

(b) Solve: $\frac{x+1}{3} - \frac{x-2}{4} = 2$ (3 marks)

(c) Solve: $x(2x-1) = 15$ (3 marks)

2. An incomplete table of values prepared for drawing the graph of the function $y = 3 - x^2$ is given below.

x	**−3**	**−2**	**−1**	**0**	**1**	**2**	**3**
y	.....	−1	2	3	2	.....	−6

(a) Complete the blanks in the table. (1 mark)

(b) Draw the graph, taking 10 small divisions along the x-axis to represent 1 unit and 10 small divisions along the y-axis to represent 1 unit. (3 marks)

(c) Using the graph,

(i) write the maximum value of the function and the coordinates of the turning point. (1 mark)

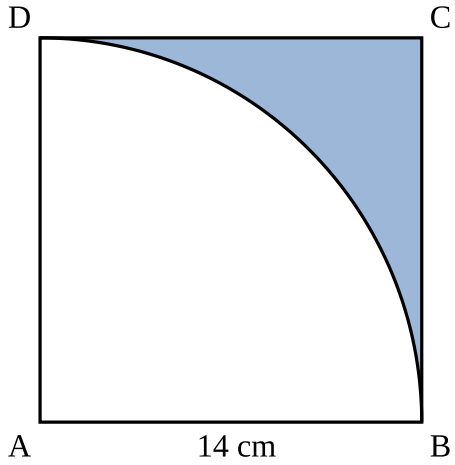
(ii) write the range of values of x for which y is positive. (1 mark)

(iii) considering $3 - x^2 = 0$, write an approximate value for $\sqrt{3}$. (2 marks)

(iv) write the value of y when $x = 2.5$. (1 mark)

(d) Write the function of the graph obtained when the graph is moved 2 units downwards along the y-axis, and write the coordinates of the points where it intersects the x-axis. (1 mark)

3. In the figure, ABCD is a square of side 14 cm. The arc BD is an arc of a circle drawn with centre A and radius AB.



(a) Find the area of the sector ABD. (2 marks)

(b) Find the area of the shaded region. (2 marks)

(c) Find the perimeter of the shaded region. (3 marks)

(d) If a semicircle is drawn outside the square with CD as its diameter, find the total area of the square and the semicircle. (3 marks)

4. In triangle ABC, AB = AC. D is the midpoint of the side BC.

(a) Prove that triangle ABD and triangle ACD are congruent. (3 marks)

(b) Show that AD $\perp$ BC. (2 marks)

(c) Show that AD bisects $B\hat{A}C$. (1 mark)

(d) If AB = 13 cm and BC = 10 cm, find the length of AD and the area of triangle ABC. (4 marks)

5. Nimal deposited a sum of Rs. 50 000 in a bank that pays an annual compound interest rate of 8%.

(a) Find the interest received at the end of the first year. (2 marks)

(b) Find the total amount in the account at the end of the second year. (2 marks)

(c) If, at the end of the second year, he withdrew Rs. 18 320 and deposited the remaining amount for one more year, find the amount in the account at the end of the third year. (3 marks)

(d) If, instead, he had deposited Rs. 50 000 for 2 years in another bank that pays 9% annual simple interest, which method is more advantageous at the end of 2 years? By how much? (3 marks)

6. Kasun saves Rs. 500 in the first month, and in each following month he saves Rs. 150 more than in the previous month.

(a) Find the amount he saves in the 10th month. (2 marks)

(b) Find the total amount he saves during the first 12 months. (3 marks)

(c) In which month does his monthly saving exceed Rs. 3000 for the first time? (3 marks)

(d) Find the total amount he saves during the first 18 months. (2 marks)

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